

TEMPORAL GRAPH RECURRENCE RESEARCH FRAMEWORK

- 1. Benchmark Calibration (Phase 0.1): SBM Regime Separation & Anti-Leakage
- 2. Continuous TGN Evaluation (Phase 1 & 1.1): Audited Recency Failure under Interference
- 3. Capacity vs Retrieval (Phase 2 & 2.1b): Response Surface & Diagnostic Injections
- 4. Generalization Suite (Phase 3): 6 Synthetic Dimensions + Real SNAP CollegeMsg
- 5. Publication Audit (Phase 4): Block-Level Statistics ($0 \notin \text{CI}(\Delta\text{AP})$), Negative Controls, & Ablations

Conclusion: Empirical Recency Failure is Addressable via Non-Parametric Historical Retrieval.